

Problem set 6.

1). $T: V \rightarrow W$, V & W are vector spaces over field F .

basis of nullspace(T): $\{\alpha_1, \dots, \alpha_k\}$

$\{\alpha_1, \dots, \alpha_{k+1}, \dots, \alpha_n\} = \text{basis}(V)$

To prove:

$\{T\alpha_{k+1}, \dots, T\alpha_n\}$ are linearly independent and span the range of T .

Proof:

Let

$$c_{k+1}T\alpha_{k+1} + \dots + c_nT\alpha_n = 0.$$

$$T(c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n) = 0.$$

now, $c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n \in V$
 $\therefore c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n = 0.$

$$T(v) = 0.$$

$$v \in \text{Nullspace}(T).$$

$$\text{Basis of Nullspace}(T) = \{\alpha_1, \dots, \alpha_k\}$$

$$\therefore v = d_1\alpha_1 + \dots + d_k\alpha_k.$$

$$\therefore d_1\alpha_1 + \dots + d_k\alpha_k = c_1\alpha_1 + \dots + c_n\alpha_n$$

$$\text{as } (\alpha_1, \dots, \alpha_n) \text{ is a basis of } V, \quad (1)$$

$$c_i = -d_i$$

$$\therefore d_1\alpha_1 + \dots + d_k\alpha_k - (c_1\alpha_1 + \dots + c_n\alpha_n) = 0$$

becomes

$$d_1\alpha_1 + \dots + d_k\alpha_k + b_1\alpha_1 + \dots + b_n\alpha_n = 0.$$

due to (1),

$$d_1 = d_k = \dots = b_1 = \dots = b_n = 0.$$

$$\therefore c_1 = c_2 = \dots = c_n = 0$$

$\therefore \alpha_1, \dots, \alpha_n$ are linearly

independent. (8)

Let $y \in \text{Range}(T)$.

So, $\exists v \in V$ such that

$$y = T(v)$$

$\{x_1, \dots, x_n\}$ is basis of V .

$$\therefore v = c_1 x_1 + \dots + c_n x_n$$

$$T(v) = c_1 T x_1 + \dots + c_n T x_n$$

But,

$$x_1, \dots, x_k \in \text{Nullspace}(T)$$

$$\therefore T x_1 = \dots = T x_k = 0.$$

$$\therefore T(v) = c_{k+1} T x_{k+1} + \dots + c_n T x_n$$

which is a linear combination of
 $\{T x_{k+1}, \dots, T x_n\}$

Hence, they span $\text{Range}(T)$.

hence proved,

2. V and W are finite dimensional vector spaces,

To prove:

- if $\dim V > \dim W$, No linear transformation from $V \rightarrow W$ is injective
- if $\dim V < \dim W$, No linear transformation from $V \rightarrow W$ is surjective.

Proof:

$$\dim(V) = \dim(\text{Null}(T))$$

$$+ \dim(\text{Range}(T))$$

a). A linear transformation is injective iff

$$\text{Null}(T) = \{0\}$$

$$\therefore \dim(\text{Null}(T)) = 0.$$

$$\therefore \dim(V) = \dim(\text{Range}(T))$$

Now, $\text{Range}(T) \subseteq W$

$$\therefore \dim(\text{Range}(T)) \leq \dim W$$

$$\therefore \dim V \leq \dim W$$

But, we need $\dim V > \dim W$.

$\therefore$ Proof by contradiction,

$\therefore$ NO SUCH injective T exists.

b) T is surjective iff $\text{Range}(T) = W$

$$\therefore \dim(\text{Range}(T)) = \dim(W)$$

$$\dim(V) = \dim(\text{Null}(T)) + \dim(W)$$

$$\dim(\text{Null}(T)) \geq 0$$

$$\therefore \dim V \geq \dim W$$

But, we need $\dim V < \dim W$.

$\therefore$ Proof by contradiction,

NO SUCH surjective T exists.

3) $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear transform

iff $\exists$ $m \times n$ matrix A

such that $T(\alpha) = A\alpha \quad \forall \alpha \in \mathbb{R}^n$.

$L(\mathbb{R}^n, \mathbb{R}^m)$ is vector space of
linear transforms from $\mathbb{R}^n \rightarrow \mathbb{R}^m$.

To find: Basis (L)

Solⁿ: Let $e_1, e_2, \dots, e_n$ be
standard basis of $\mathbb{R}^n$.

$$e_1 = (1, 0, \dots, 0), e_2 = (0, 1, \dots)$$

for $\alpha \in \mathbb{R}^n$ ~~can be~~

$$\alpha = n_1 e_1 + n_2 e_2 + \dots + n_n e_n$$

$$T(\alpha) = T(n_1 e_1 + \dots + n_n e_n)$$

$$T(\alpha) = n_1 T(e_1) + \dots + n_n T(e_n)$$

for $T(e_i)$ in $\mathbb{R}^m$,

$$A = \begin{bmatrix} | & | & & | \\ T(e_1) & T(e_2) & \dots & T(e_n) \\ | & | & & | \end{bmatrix}$$

$$\therefore A\alpha = \alpha_1 T_1(e_1) + \dots + \alpha_n T(e_n)$$

$$\therefore A\alpha = T(\alpha)$$

$$\forall \alpha \in \mathbb{R}^n.$$

Suppose there exists

$$A\alpha = \cancel{B\alpha} = T(\alpha)$$

$$B\alpha = T(\alpha)$$

$$A\alpha \neq B\alpha$$

Proof by contradiction:

$$Ae_i = T(e_i) = Be_i$$

$\therefore$ each column of A
corresponds to
a column of B

$$\therefore A = B$$

and the matrix is unique.

if $T(\alpha) = A\alpha$,

$$T(\alpha + \beta) = A(\alpha + \beta) = A\alpha + A\beta \\ = T(\alpha) + T(\beta).$$

$$T(c\alpha) = \cancel{cT(\alpha)} = A(c\alpha) \\ = c(A\alpha) \\ = cT(\alpha).$$

$\therefore T$ is linear. If $T(\alpha) = A\alpha$
for unique $m \times n$ matrix A .
Hence proved.

Let E_{ij} be a matrix

where $\cancel{n_{ij} = 1}$ $n_{ab} = 1$, if $a = i$ and $b = j$

$n_{ab} = 0$, rest

$$\therefore T_{ij}(\alpha) = E_{ij} \alpha.$$

Basis:

There are mn matrices in the set

$$\{E_{ij} : i \in [1, m], j \in [1, n]\}$$

$$\dim(L(R^n, R^m)) = mn$$

$$\text{basis}(L) = \{E_{ij} : i \in [1, m], j \in [1, n]\}$$

4). $T: V \rightarrow W$.

$\beta_1, \dots, \beta_r \in V$ and are linearly independent.
 $\beta_1, \dots, \beta_r \notin \text{Ker}(T)$.

To find: Are $T(\beta_1), \dots, T(\beta_r)$ linearly independent.

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}$

such that $T(n, y) = n$.

~~Is T linear?~~

To prove T is linear:

i). $T((n, y) + (a, b)) = T(n+a, y+b)$
 $= n+a = T(n, y) + T(a, b)$.

$$T(C(n, y)) = cn - T(\cancel{cn, y})$$

$$CT(n, y).$$

Now, if $T(n, y) = 0$
 $n = 0$

$$\therefore \text{Ker}(T) = \{ (0, y) \mid y \in R \}$$

Now, for $\beta_1 = (1, 0)$
 $\beta_2 = (1, 1)$ ~~is~~
 NOT in

$$\text{Ker}(T),$$

$$a(1, 0) + b(1, 1) = 0$$

$$\therefore a = b = 0.$$

$\therefore \beta_1$ and β_2 are linearly independent.

$$\text{Now, } T(\beta_1) = T(1, 0) = 1$$

$$T(\beta_2) = T(1, 1) = 1$$

$$\therefore a(1) + b(1) = 0,$$

$$a + b = 0$$

$$b = -a.$$

$\therefore$ for $a \in R,$

$$a(1) + (-a)(1) = 0,$$

$\therefore T(\beta_1) \dots T(\beta_r)$
 AREN'T NECESSARILY
 linearly independent.

5). V is an n -dimensional vector space over field F .

$$B = \{\alpha_1, \dots, \alpha_n\} = \text{basis}(V)$$

To find: i). $T(\alpha_j) = \alpha_{j+1} \quad j \in [1, n-1]$.
 $T(\alpha_n) = 0$.

ii). matrix representation of T in ordered basis (B)

To prove: $T^n = 0$, $T^{n-1} \neq 0$.

$$T(\alpha_1) = \alpha_2$$

$$(0, 1, 0, \dots, 0)$$

$$T(\alpha_2) = \alpha_3$$

$$(0, 0, 1, \dots, 0),$$

$$T(\alpha_n) = 0$$

$\therefore$ Matrix

$$[T]_B = \begin{bmatrix} 0 & 0 & - & - & - & 0 \\ 1 & 0 & - & - & - & 0 \\ 0 & 1 & 0 & - & - & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & - & - & - & 1 \\ 0 & 0 & - & - & - & 0 \end{bmatrix}$$

Nilpotent shift matrix.

To prove: $T^n = 0$

$$T(\alpha_j) = \alpha_{j+1}$$

$$\therefore T^2(\alpha_j) = T(\alpha_{j+1}) = \alpha_{j+2}$$

$$\therefore T^k(\alpha_j) = \alpha_{j+k}$$

$$j+k \leq n$$

$$\therefore T^n(\alpha_j) = \alpha_{j+n}$$

But $j+n > n$

$$\therefore T^n(\alpha_j) = 0$$

$$\therefore T^n(\alpha_j) = 0$$

But $j+n-1$ isn't always $> n$.

$$\therefore T^{n-1} \neq 0 \text{ as } \alpha_n \neq 0$$

Hence proved, $T^n = 0$, $T^{n-1} \neq 0$.

6). $T(x_1, x_2, x_3) = (3x_1 + x_2, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3)$

To find: matrix representation of T in standard ordered basis of $\mathbb{R}^3$.

compute matrix of T in ordered basis of $(1, 0, 1), (-1, 2, 1), (2, 1, 1)$.

$$e_1 = (1, 0, 0)$$

$$e_2 = (0, 1, 0)$$

$$e_3 = (0, 0, 1)$$

$\therefore$ columns of matrix are

$$T(e_1), T(e_2), T(e_3)$$

$$\therefore A = \begin{bmatrix} | & | & | \\ T(e_1) & T(e_2) & T(e_3) \\ | & | & | \end{bmatrix}$$

$$T(1, 0, 0) = (3, -2, 1)$$

$$T(0, 1, 0) = (1, 1, 2)$$

$$T(0, 0, 1) = (0, 0, 4)$$

$$\therefore A = [T]_{std} = \begin{bmatrix} 3 & 1 & 0 \\ -2 & 1 & 0 \\ 1 & 2 & 4 \end{bmatrix}$$

Now, Basis $V_1 = (1, 0, 1)$

$$V_2 = (-1, 2, 1)$$

$$V_3 = (2, 1, 1)$$

$$T(1, 0, 1) = (3, -2, 3)$$

~~$$T(V_1)$$~~

$$(3, -2, 3) = aV_1 + bV_2 + cV_3$$

$$= a(1, 0, 1) + b(-1, 2, 1)$$

$$+ c(2, 1, 1)$$

$\therefore$ coordinates ~~(a, b, c)~~ ,

$$(a - b + 2c, 2b + c, a + b + c)$$

$$a - b + 2c = 3$$

$$a + b + c = 3$$

$$2b + c = 2$$

$$\therefore a = 4, b = -1, c = 0$$

$$T(v_1) = 4v_1 - v_2$$

$$T(v_1) = (4, \text{---}, 0)$$

Similarly,

$$\left(\frac{9}{2}, -\frac{1}{2}, 1 \right)$$

$$T(-1, 2, 1) = (-1, 9, 9)$$

$$\therefore a - b + c = -1$$

$$a + b + c = 9$$

$$2b + c = 4$$

$$a = \frac{7}{3}, b = -\frac{8}{3}, c = \frac{20}{3}$$

$$T(v_2) = \left(\frac{7}{3}, \text{---}, \frac{20}{3} \right)$$

$$\rightarrow \left(\frac{17}{2}, \frac{7}{2}, -3 \right)$$

$$T(2, 1, 1) = (7, -3, 2)$$

$$a - b + c = 7$$

$$a + b + c = 4$$

$$2b + c = -3$$

$$\therefore a = 3, b = -1, c = 5 \quad \left(\frac{11}{2}, -\frac{3}{2}, 0 \right)$$

$$T(v_3) = (\text{---}, \text{---}, \text{---})$$

$$- [T]_{B^2}^P = \begin{bmatrix} 7/2 & 3 \\ -1 & -1/3 \\ 0 & 2/3 \end{bmatrix}$$

7). To prove:

$$\begin{bmatrix} \frac{9}{2} & \frac{17}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{7}{2} & -\frac{3}{2} \\ 1 & -3 & 0 \end{bmatrix}$$

- a). I is only matrix similar to I.
 b). O is only matrix similar to O.
 c). A and B are similar.
 A is invertible iff B is invertible.
 d). Similarity is an equivalence relation.

Proof: A is similar to B if
 $\exists$ invertible P such that

$$B = P^{-1}AP.$$

a). $B = P^{-1}IP$

Now, $IP = P$

$$B = P^{-1}P$$

$$P^{-1}P = I$$

$$\therefore \boxed{B = I}$$

Hence proved.

b) $B = P^{-1}(0)P$

$$(0)P^{-1} \Rightarrow 0$$

$$B = P^{-1}(0)$$

$$P^{-1}(0) \Rightarrow 0$$

$$\therefore \boxed{B \Rightarrow 0}$$

Hence proved, only 0 is similar to zero matrix.

c) $B = P^{-1}AP$

→ proof:

A is invertible.

$\therefore \exists A^{-1}$ such that

$$A^{-1}A = AA^{-1} = I$$

Now,

$$B^{-1} = (P^{-1}AP)^{-1}$$

$$B^{-1} = P^{-1}A^{-1}(P^{-1})^{-1}$$

$$\boxed{B^{-1} = P^{-1}AP}$$

Hence, as P^{-1} , A and P
exist,
 $P^{-1}AP$ exists
 $\therefore B$ exists.

← proof:

B is invertible.

$\therefore \exists B^{-1}$ such that

$$B^{-1}B = BB^{-1} = I.$$

$$\therefore B = P^{-1}AP$$

$$PB = PP^{-1}AP = AP$$

$$\boxed{PBP^{-1} = A}$$

As P, B, P^{-1} are invertible,
 A is invertible.

Hence proved.

A is invertible iff B is invertible
for similar matrices A & B .

d). i). Reflexive:

$$\text{for } A, \quad A = I^{-1} A I$$

$\therefore A$ is similar to itself
 $\& \therefore I$ is invertible.

$\therefore$ reflexive.

ii). Symmetric.

for A similar to B

$$A = P^{-1} B P$$

$$\therefore P A = P P^{-1} B P$$

$$P A = B P$$

$$P A P^{-1} = B$$

Now, for $Q = P^{-1}$,

$$Q^{-1} A Q = B$$

$\therefore B$ similar to A .

$\therefore$ Symmetric.

iii), Transitive:

for A similar to B ,
 B similar to C .

$$A = P^{-1} B P$$

$$B = Q^{-1} C Q$$

$$\therefore A = (P^{-1} Q^{-1}) (C) (Q P)$$

Let $Q P$ be matrix R

$$\therefore (Q P)^{-1} = (P^{-1} Q^{-1}) = R^{-1}$$

$$\therefore \boxed{A = R^{-1} C R}$$

$\therefore A$ is similar to C .

~~It~~ $\therefore$ transitive.

Hence, it's an equivalent
relation.